THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 9 — SCIENCE · MEDIUM LEVEL UNIT TEST SET 3

Chapters 9–13: **Cell: The Building Block of Life, Tissues in Action, Patterns in Life: Diversity and Classification, Reproduction: How Life Continues, Earth as a System: Energy, Matter and Life** | RBSE Pattern | Session 2026–27

Time: 2 Hours Name: _____ Roll No.: _____ **Maximum Marks: 50**

General Instructions:

1. All questions are compulsory.
2. Q. 1–15 are Objective Type (1 mark each).
3. Q. 16–20 are Very Short Answer Type (2 marks each).
4. Q. 21–25 are Short Answer Type (3 marks each).
5. Q. 26–27 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 24–25 and Q. 26–27.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 15 × 1 = 15

Choose the correct option and write it in your answer book.

1. Which organelle is absent in animal cells? 1

(a) mitochondrion **(b)** chloroplast **(c)** Golgi **(d)** ribosome

2. The structure that reduces friction at joints is: 1

(a) bone **(b)** tendon

(c) cartilage and synovial fluid **(d)** muscle

3. Two species placed in the same genus are: 1

(a) more closely related than species in the same family but different genera

(b) identical

(c) of different kingdoms

(d) unrelated

4. A cell of the human body has 46 chromosomes. After mitosis each daughter cell has: 1

(a) 23 **(b)** 44 **(c)** 46 **(d)** 92

5. The forest absorbs CO2 at 20 tonnes per hectare per year. For 50 hectares per year it absorbs: 1

(a) 400 tonnes **(b)** 2.5 tonnes **(c)** 70 tonnes **(d)** 1000 tonnes

6. The cell division that increases the number of cells for growth is: 1

(a) meiosis **(b)** fertilisation **(c)** osmosis **(d)** mitosis

7. Annual rings in a tree trunk are formed in: 1

(a) phloem **(b)** secondary xylem **(c)** cork **(d)** pith

8. The seed of wheat is a: 1

(a) monocot seed **(b)** dicot seed **(c)** gymnosperm seed **(d)** spore

9. The ovulation takes place on about: 1

(a) the 14th day of a 28-day cycle **(b)** the 7th day

(c) the 28th day **(d)** the 1st day

10. The movement of water from land to sea through rivers is: 1

(a) evaporation **(b)** run-off **(c)** condensation **(d)** transpiration

11. Which cell structure would be most affected if a cell has no ribosomes? 1

(a) protein synthesis **(b)** cell division
(c) respiration only **(d)** photosynthesis

12. A tissue section shows living cells with unevenly thickened corners. The tissue is: 1

(a) collenchyma **(b)** parenchyma **(c)** xylem **(d)** sclerenchyma

13. An organism has a notochord, a dorsal nerve cord and gill slits at some stage. It is a: 1

(a) arthropod **(b)** annelid **(c)** mollusc **(d)** chordate

14. A student sees tiny buds on the margin of a Bryophyllum leaf. They are used for: 1

(a) storage **(b)** photosynthesis
(c) respiration **(d)** vegetative reproduction

15. Lightning contributes to the nitrogen cycle by: 1

(a) making CO2 **(b)** converting N2 into nitrogen oxides
(c) forming proteins **(d)** making ozone

SECTION — B (Very Short Answer Type) 5 × 2 = 10

Answer in about 20–30 words each.

16. State two differences between rough and smooth ER. 2

17. State two differences between bone and cartilage. 2

18. How do you distinguish a roundworm from an earthworm? 2

19. Why do the seeds need dormancy? Give one reason. 2

20. Why are the polar regions more affected by climate change? 2

SECTION — C (Short Answer Type) 5 × 3 = 15

Answer in about 40–60 words each.

21. Describe how plastics affect the Earth system. 3

22. Explain how the surface area to volume ratio limits the size of cells and why a big organism has many small cells. 3

23. Differentiate between simple and complex tissues with two examples each. 3

24. Classify the following animals into phyla: earthworm, hydra, octopus, tapeworm, starfish, cockroach. 3

OR

Explain how fertilisation takes place in a flowering plant after pollination. 3

25. Explain how the ozone hole forms over Antarctica in brief. 3

OR

The cell of a cow and that of a mouse are of similar sizes. How does the organism size differ? 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

26. Explain the structure of seeds (bean and maize), the parts, and the process of germination. 5

OR

Calculate the amount of CO2 absorbed by a forest of 100 hectares if each hectare absorbs 6 tonnes per year; compare it with the emission of 600 cars each emitting 2.4 tonnes per year. 5

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27. Explain how the movement of substances across the cell membrane occurs: diffusion, osmosis and active transport with examples. 5

OR

Describe the structure of the xylem and phloem using a transverse section, the position of the two in a vascular bundle in the root and the stem. 5

END OF QUESTION PAPER **Total: 50 Marks**

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